MARYAM REZAEE

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A computer science researcher focused on the intersection of LLM Interpretability and Cognitively-Inspired AI. My work centres on opening the black box of generative models using concept-level attribution and energy-based frameworks to verify generation logic. I aim to bridge this technical expertise with cognitive modelling, exploring how Neurosymbolic architectures with human inductive biases can mitigate hallucinations and enforce logical consistency. My objective is to build transparent, human-aligned systems that combine the fluency of neural networks with the structural rigour of symbolic reasoning.

Education

M.Sc. in Computer Science — Sep 2023 – (Expected) Aug 2026

Department of Mathematical Sciences, Sharif University of Technology, Iran — AVG: 18.57 / 20.00

Thesis: *Interpretability in Generative Models: Investigating the Mechanisms Behind Output Generation in Large Language Models*

B.Sc. in Computer Science — Sep 2019 – Sep 2023

Department of Mathematical Sciences, University of Isfahan, Iran — AVG: 16.66 / 20.00

Project: *Understanding the Application of Machine Learning in Cell Signalling Pathway Analysis*

Research Experience

Exploring Neurosymbolic AI and Cognitive Architectures — Jul 2025 – Present

Independent Researcher

Sharif University of Technology, Tehran, Iran

- Conducting a self-directed literature review on the integration of neural networks with symbolic reasoning systems to inform future research.
- Studying key topics including program synthesis, logic-based model integration, Computational Narratology, and the role of human inductive biases (specifically Active Inference and the FEP) in cognitive architectures.
- Further developed a novel “Director-Actor” Neurosymbolic architecture that integrates symbolic planning with neural generation to minimise predictive error and solve the “wandering plot” problem in LLMs.
- Implemented a seq2seq model for program generation as a practical exploration of neurosymbolic principles.

Concept-Based Interpretability for Retrieval-Augmented Generation (RAG) Systems on Large Language Models — May 2025 – Present

Research Mentor under the supervision of Dr. Fatemeh Seyyedsalehi

Trustworthy and Generative Machine Learning (TGML) Lab, Sharif University of Technology, Tehran, Iran

- Mentoring a B.Sc. student in developing a novel framework for RAG interpretability.
- Investigating the application of Concept Bottleneck Models (CBMs) to explain the internal generation processes.
- Developing methods using Concept Activation Vectors (CAV) and Automatic Concept-based Explanations (ACE) to enhance the transparency and traceability of RAG outputs.

A Concept Level Energy-Based Framework for Interpreting Black-Box Large Language Model Responses

Feb 2025 – Present Link

Graduate Researcher under the supervision of Dr. Fatemeh Seyyedsalehi
TGML Lab, Sharif University of Technology, Tehran, Iran

- Proposed a model-agnostic, post-hoc framework (ESCI) to interpret black-box LLMs accessed only via APIs.
- Designed an energy model to quantify the conceptual links between user prompts and LLM-generated responses.
- Trained an efficient interpreter network to provide sentence-level importance scores.
- Established proof-of-concept results and currently conducting extensive experimental validation to refine the methodology for submission to ACL; for preliminary preprint, view the above link.

Interpretability in Generative Models: Investigating the Mechanisms Behind Output Generation in Large Language Models

Sep 2024 – Present

Master's Thesis Researcher under the supervision of Dr. Fatemeh Seyyedsalehi
TGML Lab, Sharif University of Technology, Tehran, Iran

- Conducted a comprehensive review of intrinsic, post-hoc, concept-based, & mechanistic interpretability methods.
- Developed a post-hoc framework to interpret instance-based input-output attribution in LLM generation.
- Analysing internal model mechanisms and their influence on output generation.
- Investigating ways to leverage interpretability findings to improve model robustness, alignment, and performance.

Understanding the Application of Machine Learning in Cell Signalling Pathway Analysis and Developing a DEG Analysis Handbook

Feb 2023 – Sep 2023 Link

Bachelor's Project Researcher under the supervision of Dr. Fatemeh Mansoori
University of Isfahan, Isfahan, Iran

- Conducted a literature review on the application of ML and DL (e.g., SVMs) in bioinformatics.
- Analysed and documented methodologies for Gene Expression and Differentially Expressed Gene (DEG) analysis.
- Authored a technical handbook and implementation roadmap for future students to use in DEG analysis.

A Neurosymbolic Architecture for Coherent Narrative Generation

Jan 2021 | 2025 Link

Undergraduate Researcher under the supervision of Dr. Jaafar Almasizadeh
University of Isfahan, Isfahan, Iran

- Designed a Neurosymbolic "Director-Actor" architecture to improve LLM coherence, operationalising narrative structures via Signal Processing (Gaussian envelopes), Gated FSMs, and A* Search.
- Modelled character archetypes as Vector Centroids to enforce consistency and framed narratives as cognitive data compression schemas optimising for Active Inference.

Foundational Research in Cognitive Neuroscience

Aug 2016 – Feb 2017

Team Researcher (1st Kashan Neuroscience Workshop for Students)
Kashan University of Medical Sciences, Kashan, Iran

- Achieved 2nd Place in a multi-stage research competition for a portfolio on neuroscience and psychology.
- Researched and presented on cognitive topics, including the neural basis of creativity (divergent thinking), hypnotic analgesia, and psychological models of learning/cognitive styles.
- Conducted foundational reviews on core neuroanatomy (Limbic System), the brain-mind problem, and applied clinical neuroscience as part of the workshop curriculum.

Three Centuries in Pursuit of the Theory of Everything

Oct 2015 – Feb 2016

Independent Researcher (Khawrazmi Youth Award)
Farzanegan School (NODET: National Organization for Development of Exceptional Talents), Kashan, Iran

- Awarded 2nd Place in the Physics category at the Khawrazmi Youth Award (Kashan regional).
- Authored a comprehensive research paper on the historical and theoretical development of a unified field theory in physics, analysing key milestones from classical mechanics to modern physics.

Teaching Experience

Relevant to Academic Major

- **Head Teaching Assistant: *Generative Models*** *(M.Sc.)* — Sep 2025 – Present
  Dr. Fatemeh Seyyedsalehi, Sharif University of Technology
- **Teaching Assistant: *Machine Learning Operations*** *(B.Sc.)* — Sep 2025 – Present
  Dr. Fatemeh Seyyedsalehi, Sharif University of Technology
- **Head Teaching Assistant: *Machine Learning*** *(B.Sc.)* — Sep 2025 – Present
  Dr. Mohsen Alambardar, University of Isfahan
- **Instructor: *Fundamentals of Technological Product Development*** — Jul 2024 – Sep 2024
  Tehran University of Medical Sciences
- **Head Teaching Assistant: *Advanced Programming*** *(B.Sc. Math & CS)* — Jan 2023 – Jun 2023
  Dr. Fatemeh Mansoori, University of Isfahan
- **Head Teaching Assistant: *Operating Systems*** *(B.Sc.)* — Jan 2023 – Jun 2023
  Dr. Mojtaba Rafiee, University of Isfahan
- **Instructor: *Algorithms and Data Structures*** *(B.Sc.)* — Nov 2022 – Feb 2023
  Private Classes, University of Isfahan
- **Instructor: *Python Programming*** *(Elementary to Advanced)* — Sep 2019 – Feb 2023
  Private Classes, University of Isfahan

Additional Topics

- **Instructor and Mentor: *Graphic Design with Adobe Illustrator*** — Jan 2023 – Jun 2023
  Academic Association of Mathematics and Computer Science (AMCSUI), University of Isfahan
- **Instructor: *English Language*** *(Intermediate to Advanced)* — Sep 2016 – Nov 2023
  Private Classes

Selected Course Projects

- **System 2 AI** (M.Sc.) — Feb 2025 – Jun 2025
  Dr. M.H. Rohban & Dr. M. Soleymani, Sharif University of Technology — Audit
  PY Program Generation w/ Seq2Seq Models | PY Symbolic Regression | PY GraphRAG | PY LLM Agent
- **Deep Learning** (M.Sc.) — Feb 2025 – Jun 2025
  Dr. Fatemeh Seyyedsalehi, Sharif University of Technology — Mark: 20.00 / 20.00
  PY GPT2 | PY Math Reasoning in LLMs | PY PINN | PY Atari DQN | PY SSL & Contrast. Learning | PY FNN from Scratch

Generative Models (M.Sc.) Sep 2024 – Feb 2025
Dr. Fatemeh Seyyedsalehi, Sharif University of Technology Mark: 18.00 / 20.00
PY RNN vs Transformer | PY Diffusion/DDPM | PY EBM | PY β-VAE | PY Pix2Pix | PY Norm. Flows | PY Bayesian Network

Machine Learning (M.Sc.) Sep 2024 – Feb 2025
Dr. Ali Sharifi Zarchi, Sharif University of Technology Mark: 20.00 / 20.00
PY Neural Decompo. | PY LoRA from Scratch | PY Word2Vec | PY Knowl.Distil. (KD) & CLIP | PY MobileNet, Eff. & KD.

Matrix Computations (M.Sc.) Sep 2023 – Feb 2024
Dr. Mohammad Reza Razvan, Sharif University of Technology Mark: 19.00 / 20.00
PY SVD RGB-Image Compressor from Scratch | MATLAB Comparing Linear Solvers Implemented from Scratch

Computer Networking (B.Sc.) Feb 2023 – Jun 2023
Dr. Mohsen Alambardar, University of Isfahan Mark: 18.75 / 20.00
PY Group Chat Application with TCP Protocol and Highspeed Data Sharing

Artificial Intelligence (B.Sc.) Sep 2022 – Feb 2023
Dr. Fatemeh Mansoori, University of Isfahan Mark: 19.50 / 20.00
PY AI Search Algorithms for Solving Mazes | PY MDP Gridworld Solver with Value & Policy Iteration

Cryptography (B.Sc.) Sep 2022 – Feb 2023
Dr. Mojtaba Rafiee, University of Isfahan Mark: 19.50 / 20.00
Feasibility of Proving Authenticity and Data Integrity for P2P Using Hash

Fundamentals of Operating Systems (B.Sc.) Feb 2022 – Jun 2022
Dr. Mojtaba Rafiee, University of Isfahan Mark: 19.50 / 20.00
PY Reader-Writer Lock Implementation with Three Priorities

Mathematical Databases (B.Sc.) Feb 2022 – Jun 2022
Dr. Fatemeh Mansoori, University of Isfahan Mark: 19.60 / 20.00
PostgreSQL University Course Registration System | PostgreSQL Academic Transcript Generator & Analyzer

Mathematical Software (B.Sc.) Feb 2022 – Jun 2022
Dr. Najmeh Hoseini, University of Isfahan Mark: 17.75 / 20.00
MATLAB Audio-Signal Analysis | MATLAB CLI University Portal for Grade Management

Algorithms & Data Structures (B.Sc.) Feb 2022 – Jun 2022
Dr. Jaafar Almasizadeh, University of Isfahan Mark: 20.00 / 20.00
PY Chinese Postman | PY Huffman Compressor | PY Max-Flow Altered | PY Prime Sieves Anal. | PY Topol. Sort Anal.

Advanced Programming (B.Sc.) Feb 2020 – Jun 2020
Dr. Jaafar Almasizadeh, University of Isfahan Mark: 20.00 / 20.00
PY RSA Cryptosystem Implementation from Scratch | PY CLI Banking System

Selected Talks

- **ReAGent: A Model-Agnostic Feature Attribution Method for LMs** Link Aug 2025
  Trustworthy & Generative ML (TGML) Lab, Sharif University of Technology
- **Interpreting Language Models with Contrastive Explanations** Link Jun 2025
  TGML Lab, Sharif University of Technology
- **Propagating Universal Perturbations to Attack LLM Guard-Rails** Link May 2025
  Deep Learning Seminars, Sharif University of Technology
- **Universal and Transferable Adversarial Attacks on Aligned LMs** Link May 2025
  Deep Learning Seminars, Sharif University of Technology
- **An Exploration of Concept-Based Interpretability Methods** Link Mar 2025
  TGML Lab, Sharif University of Technology
- **DiT: Scalable Diffusion Models with Transformers** Link Feb 2025
  Generative Models Seminars, Sharif University of Technology
- **Saliency-Guided Training: Interpretability and Robustness in DNNs** Link Jan 2025
  Machine Learning Seminars, Sharif University of Technology
- **Model Sparsity Can Simplify Machine Unlearning** Link Dec 2024
  Machine Learning Seminars, Sharif University of Technology
- **A Foray into Bioinformatics: ML in Cell Signalling Pathway Analysis** Link Jun 2023
  Bachelor's Project Presentation, University of Isfahan
- **Key Takeaways from the 19th Iranian Conference on HPE** May 2018
  Invited Speaker, Post-Conference Seminar, Kashan University of Medical Sciences

Technical Skills

Areas of Expertise	Languages / Core Tools	Modules / Frameworks	Technical Software
ML/DL	Python	PyTorch	Illustrator
GMs/LLMs	R	TensorFlow	InDesign
XAI	MATLAB	Scikit-learn	Photoshop
NeSy AI	PostgreSQL	NumPy	Premiere Pro
Reasoning	LaTeX	Pandas	After Effects
RAG	JavaScript	Transformers	Audition
MLOps	HTML/CSS	Diffusers	Figma
Bioinform	Sockets	Threading	Canvas
CogNeuro	Git/Bash	RISC-V	Docker

Leadership & Project Management

Conceptual Prototype Designer Jul 2024 – Aug 2024
"Siba," the Interprofessional Gamified Platform for Intelligent Education and Evaluation of Medical, Nursing, and Pharmacy Students | *16th Scientific Olympiad for Medical Science Students of Iran*

Founder, Manager, Lead Designer May 2023 – Mar 2024
"PsyCity," the National Competetive and Educational Game for Students of Computer Science

Committee Representative Apr 2023 – May 2024
Assembly for the Academic Associations | *University of Isfahan*

Association President, Head of Content Creation Branch Oct 2022 – May 2024
Association for Mathematics & Computer Science (AMCSUI) | *University of Isfahan*

Content Lead Mar 2022 – Apr 2022
"Varpharin," the Gamified Event for Diabetic Patient Management with an Interprofessional Approach | *AMEE Grant Recipient, Presented at AMEE 2023 (Short Communication)*

Professional Activities & Service

Invited AI Panelist Aug 2025
Symposia on AI in Health Professions Education | *International AMEE 2025 Conference, Live*

English Interpreter & Student Task Force May 2023
International Guests & Presenters | *24th Iranian Conference on Health Professions Education*

UI/UX & Graphic Designer Since 2018
Academic & Institute Projects | *Freelance*

Academic & Technical Translator Since 2016
Book & Article Publication | *Freelance*

Awards

Honouree: **Tech & Innovation** 2024
17th International Harekat Festival, Iran

Top: **Best & Most Creative Pres.** 2024
17th University-Level Harekat Festival, UI, Iran

Honouree: **Educational Assoc.** 2023
16th International Harekat Festival, Iran

2nd: **Best Research in Physics** 2016
18th Khawrazmi Youth Award, Kashan, Iran

Top: **Digital Content** 2024
17th International Harekat Festival, Iran

1st: **Most Active Association** 2024
Tadaee Festival, University of Isfahan, Iran

Honouree: **Best Team** 2018
Avicenna Student Creativity Cup, Iran

1st: **Poetry** 2016, 2017, 2018
Regional Cultural-Artistic Cup, Kashan, Iran

Top: **Best Science Assoc.** 2024
17th University-Level Harekat Festival, UI, Iran

Honouree: **Tech & Innovation** 2024
Roshana Festival, Science Ministry, Iran

2nd: **Best Team** 2016
1st Kashan Neuroscience Workshop, Iran

1st: **Short Story** 2016, 2017
Regional Cultural-Artistic Cup, Kashan, Iran

Languages

— Full Proficiency in ***English***

— Native Fluency in ***Persian (Farsi)***